

Delayed Smoothing Defeats Global One-Sided Resolvent Propagation

Independent research note

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Abstract

Arora and Glück ask whether a one-sided rank-one estimate for a resolvent at one real point propagates to every real point of the resolvent set when their general domain-smoothing exponents are used. We give a negative answer to both the lower- and upper-estimate versions. On three copies of $L^2(\mathbb{T})$, a positive cyclic operator transfers two components unchanged and applies Poisson smoothing on the third transition. Its cube, and the cube of its adjoint, have the required smoothing, but moving the resolvent parameter to the forbidden side exposes a negative identity channel. Indicator functions of shrinking arcs show that this channel cannot be bounded by the prescribed rank-one operator. The example satisfies the source hypotheses with $m_1 = m_2 = 3$.

Packet status

Run	fa_banach_001, lane 13.
Source	Arora–Glück, arXiv:2104.12205, question after Proposition 4.7 on PDF page 15 [1].
Classification	Candidate counterexample, likely valid subject to human review.
Answer	Neither order restriction in Theorem 4.5 can be removed in the generality of Setting 1.1.

The situation becomes a bit simpler than in Theorem 4.5 if the domination condition in Setting 1.1 is satisfied for the exponents $m_1 = m_2 = 1$:

Proposition 4.7. *Assume that the domination condition in Setting 1.1 is satisfied for $m_1 = m_2 = 1$. Let μ_0 be a real number in $\rho(A)$.*

- (a) *We have $\mathcal{R}(\mu_0, A) \succeq -u \otimes \varphi$ if and only if $\mathcal{R}(\mu, A) \succeq -u \otimes \varphi$ for all real numbers μ in the resolvent set of A .*
- (b) *We have $\mathcal{R}(\mu_0, A) \preceq u \otimes \varphi$ if and only if $\mathcal{R}(\mu, A) \preceq u \otimes \varphi$ for all real numbers μ in the resolvent set of A .*

Proof. Let $\mu \in \rho(A) \cap \mathbb{R}$. Corollary 2.6 gives us $-u \otimes \varphi \preceq \mathcal{R}(\mu, A)\mathcal{R}(\mu_0, A) \preceq u \otimes \varphi$. Therefore, the resolvent equation

$$\mathcal{R}(\mu, A) = \mathcal{R}(\mu_0, A) + (\mu_0 - \mu)\mathcal{R}(\mu, A)\mathcal{R}(\mu_0, A).$$

immediately implies the proposition. $\square$

Thus, if we want to derive a one-sided estimate for $\mathcal{R}(\mu, A)$ from a one-sided estimate for $\mathcal{R}(\mu_0, A)$, Proposition 4.7 tells us that, if $m_1 = m_2 = 1$, then we do not need to care whether $\mu \leq \mu_0$ or $\mu \geq \mu_0$. For the more general m_1 and m_2 , we do not know whether the conditions $\mu \leq \mu_0$ and $\mu \geq \mu_0$ in Theorem 4.5(a) and (b), respectively, can be dropped.

5. EVENTUAL POSITIVITY OF RESOLVENTS AND THEIR POWERS

In this section, we finally come back to sufficient criteria – and characterizations – of uniform (anti-)maximum principles. Our goals in this section are to finally prove Theorem 1.2, to give a similar result on powers of the resolvent (Theorem 5.2) albeit

1 The source question

Let A be a closed, densely defined real operator on a complex Banach lattice E . The source fixes $u \in E_+$ and a strictly positive $\varphi \in E'_+$, and assumes that, for some nonnegative integers m_1, m_2 ,

$$\text{dom}(A^{m_1}) \subseteq E_u, \quad \text{dom}((A')^{m_2}) \subseteq (E')_\varphi. \quad (1)$$

It also imposes a positive spectral hypothesis at a real eigenvalue. Writing $P = u \otimes \varphi$ and $\mathcal{R}(\mu, A) = (\mu - A)^{-1}$, Theorem 4.5 proves

$$\begin{aligned} \mathcal{R}(\mu_0, A) \succeq -P, \mu \leq \mu_0 &\implies \mathcal{R}(\mu, A) \succeq -P, \\ \mathcal{R}(\mu_0, A) \preceq P, \mu \geq \mu_0 &\implies \mathcal{R}(\mu, A) \preceq P. \end{aligned} \quad (2)$$

Here $S \succeq -P$ means $S \geq -cP$ for some $c > 0$, and similarly for $\preceq$. Proposition 4.7 removes the restrictions on μ when $m_1 = m_2 = 1$. The source asks whether they can also be dropped for general m_1, m_2 [1, p. 15].

Idea of the proof

The source assumptions only require eventual smoothing by powers of the operator and its adjoint. We therefore place two unsmoothed identity transfers before one smoothing step in a three-phase cycle. Cubing the cycle gives Poisson smoothing in every phase, so (1) holds with $m_1 = m_2 = 3$. At the initial resolvent point the cycle itself is positive. But the exact resolvent identity on the other side begins with $Q - \delta Q^2$; the $-\delta Q^2$ term transports one component unchanged into a phase where the Q term vanishes. A rank-one integral operator sees an indicator of a small arc only through its small measure, whereas this negative identity channel retains pointwise size one. This mismatch violates every possible rank-one lower bound. Changing A to $-A$ gives the upper-bound failure.

2 The delayed-smoothing operator

Give $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ normalized Lebesgue measure. Fix $r \in (0, 1)$, for example $r = 1/2$, and let K be convolution on $L^2(\mathbb{T})$ with the Poisson kernel

$$p_r(x) = \frac{1 - r^2}{1 - 2r \cos(2\pi x) + r^2}, \quad \widehat{p_r}(n) = r^{|n|}. \quad (3)$$

Thus K is positive, self-adjoint, injective, has dense range, satisfies $K\mathbf{1} = \mathbf{1}$, and has no other fixed vectors. Since $p_r \in L^2(\mathbb{T})$, Young's inequality gives $K : L^2(\mathbb{T}) \rightarrow L^\infty(\mathbb{T})$ boundedly.

Set

$$E = L^2(\mathbb{T})^3, \quad \varphi(f) = \frac{1}{3} \sum_{j=0}^2 \int_{\mathbb{T}} f_j, \quad u = (\mathbf{1}, \mathbf{1}, \mathbf{1}),$$

and use the coordinatewise lattice order. Then φ is strictly positive and $E_u = L^\infty(\mathbb{T})^3$ (as a set). Define

$$Q(f_0, f_1, f_2) = (f_1, f_2, Kf_0). \quad (4)$$

Lemma 1. *Q is a bounded positive real operator, is injective, and has dense range. Moreover,*

$$Q^3 = \text{diag}(K, K, K), \quad Q'(g_0, g_1, g_2) = (Kg_2, g_0, g_1), \quad (Q')^3 = \text{diag}(K, K, K). \quad (5)$$

The fixed spaces of both Q and Q' consist precisely of the global constant triples $(c\mathbf{1}, c\mathbf{1}, c\mathbf{1})$.

Proof. Positivity and (5) follow directly from (4). Injectivity follows from that of K . The range is $L^2(\mathbb{T}) \times L^2(\mathbb{T}) \times \text{ran } K$, which is dense because K is self-adjoint and injective. If $Qf = f$, the first two coordinates give $f_0 = f_1 = f_2$, and the last gives $Kf_0 = f_0$; (3) then makes f_0 constant. The same calculation with the displayed formula for Q' proves the adjoint assertion. $\square$

Because Q is injective with dense range, its inverse is a closed densely defined operator. Put

$$A = I - Q^{-1}, \quad \text{dom } A = \text{ran } Q. \quad (6)$$

Lemma 2. *A satisfies all domination and spectral hypotheses of Setting 1.1 in the source with $m_1 = m_2 = 3$, u and φ as above, and $\lambda_0 = 0$.*

Proof. A is closed, densely defined, and real. If C is a closed operator, then induction gives

$$\text{dom}((I - C)^k) = \text{dom}(C^k) \quad (7)$$

for every positive integer k : once $x \in \text{dom } C$, membership of $(I - C)x$ in $\text{dom}(C^{k-1})$ is equivalent, by linearity and induction, to membership of Cx there. With $C = Q^{-1}$, (5)–(7) yield

$$\text{dom}(A^3) = \text{dom}(Q^{-3}) = \text{ran}(Q^3) \subseteq L^\infty(\mathbb{T})^3 = E_u. \quad (8)$$

The standard inverse-adjoint identity gives $(Q^{-1})' = (Q')^{-1}$, and the same argument using $(Q')^3$ proves

$$\text{dom}((A')^3) \subseteq L^\infty(\mathbb{T})^3 = (E')_\varphi. \quad (9)$$

Finally, $Af = 0$ is equivalent to $Qf = f$, so Lemma 1 makes zero a geometrically simple eigenvalue with eigenvector u . Its adjoint fixed-space statement gives the required eigenfunctional, represented by the constant one and hence equal to φ up to normalization. Thus the required strong comparisons with u and φ hold as equalities. $\square$

3 Failure of the lower estimate

Let $P = u \otimes \varphi$. Since

$$1 - A = Q^{-1}, \quad \mathcal{R}(1, A) = Q \geq 0 \geq -P, \quad (10)$$

the premise of the lower-estimate question holds at $\mu_0 = 1$.

Lemma 3. *For every $\delta > 0$, the number $1 + \delta$ belongs to $\rho(A)$, but*

$$\mathcal{R}(1 + \delta, A) \not\subseteq -P. \quad (11)$$

Proof. The polynomial identity

$$(I + \delta Q)(I - \delta Q + \delta^2 Q^2) = I + \delta^3 Q^3 \quad (12)$$

and (5) show that $I + \delta Q$ is invertible, because $I + \delta^3 K$ is invertible. From (6) and (12),

$$\mathcal{R}(1 + \delta, A) = Q(I + \delta Q)^{-1} = (Q - \delta Q^2 + \delta^2 Q^3)(I + \delta^3 Q^3)^{-1}. \quad (13)$$

Write $H = (I + \delta^3 K)^{-1}$. For $h \in L^2(\mathbb{T})$, (13) gives

$$\mathcal{R}(1 + \delta, A)(0, 0, h) = (-\delta Hh, Hh, \delta^2 K Hh). \quad (14)$$

The operator

$$M = \delta^3 K(I + \delta^3 K)^{-1}$$

is convolution with a bounded continuous kernel: its Fourier coefficients are

$$\frac{\delta^3 r^{|n|}}{1 + \delta^3 r^{|n|}},$$

which are absolutely summable. Let C be the supremum norm of this kernel. Since $H = I - M$, for every measurable arc $B \subset \mathbb{T}$ and almost every $x \in B$,

$$H\mathbf{1}_B(x) \geq 1 - C|B|. \quad (15)$$

Suppose that $\mathcal{R}(1+\delta, A) \geq -cP$ for some $c > 0$. Apply this operator inequality to the positive vector $(0, 0, \mathbf{1}_B)$. Its P -image is the global constant $|B|/3$, whereas (14)–(15) show, whenever $C|B| \leq 1/2$, that the first component of its resolvent image is at most $-\delta/2$ on B . The assumed lower bound would force

$$-\frac{\delta}{2} \geq -\frac{c|B|}{3}.$$

Choosing a positive-length arc with $|B| < \min\{1/(2C), 3\delta/(2c)\}$ contradicts this inequality. No c exists, which proves (11). $\square$

4 Answer to both parts

Theorem 1 (negative answer to the source question). *In the generality of Setting 1.1 of Arora–Glück, neither the condition $\mu \leq \mu_0$ in Theorem 4.5(a) nor the condition $\mu \geq \mu_0$ in Theorem 4.5(b) can be dropped. This already fails with $m_1 = m_2 = 3$ on the Hilbert lattice $L^2(\mathbb{T})^3$.*

Proof. Lemmas 2 and 3, together with (10), give the counterexample to part (a): the lower estimate holds at $\mu_0 = 1$ but fails at every $\mu = 1 + \delta > \mu_0$.

For part (b), replace A by $B = -A$. The domination and spectral hypotheses are unchanged. At $\nu_0 = -1$,

$$\mathcal{R}(-1, B) = -\mathcal{R}(1, A) = -Q \leq 0 \leq P,$$

so the upper estimate holds. At $\nu = -1 - \delta < \nu_0$,

$$\mathcal{R}(-1 - \delta, B) = -\mathcal{R}(1 + \delta, A).$$

If this were bounded above by a positive multiple of P , then $\mathcal{R}(1 + \delta, A)$ would be bounded below by a negative multiple of P , contradicting Lemma 3. $\square$

5 Verification, limitations, and novelty

The accompanying script discretizes $\mathbb{T}$ by cyclic grids of sizes 12, 24, 48, 96, 192. It verifies (5) and (13) to machine precision and computes the least constants in the finite-dimensional inequalities $S \geq -cP$. They are approximately

$$17.473, 35.473, 71.473, 143.473, 287.473,$$

and their ratio to grid size tends to 1.5. This is consistent with, but is not a substitute for, the shrinking-arc proof.

The construction is an abstract Banach-lattice operator, not a differential operator. That is enough to answer the source’s question in its stated general setting, but it does not show that the side restrictions are necessary for any narrower elliptic or semigroup class. It also does not determine the minimal smoothing exponents at which failure can first occur.

On August 17, 2026, the run’s four lightweight indexes were searched by the arXiv id, exact title, and core terms. Bounded web searches used the exact question, theorem terminology, title, authors, and variants of one-sided rank-one resolvent domination. They found the source and related later work on eventual domination and individual maximum principles, but no explicit resolution or this construction. This is a bounded novelty check, not an exhaustive priority claim.

The recommended human-review focus is (7)–(9) and the conversion of (14)–(15) into failure of operator order. The remaining phase algebra and the sign reversal are direct.

References

- [1] S. Arora and J. Glück, *An operator theoretic approach to uniform (anti-)maximum principles*, J. Differential Equations **310** (2022), 164–197; arXiv:2104.12205.